IN-APP COMMUNICATION MODULE

SmartCampus School Management System
Technical Specification & Implementation Guide — v1.0 · March 2026
Confidential · Internal Documentation

Audience: Development Team · School Administrators · IT / Operations Team

Related Modules:
Student Management · Staff Management · Attendance · HR & Payroll · Finance · Parent Portal · Mobile App

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1. MODULE OVERVIEW

1.1 Purpose

The **In-App Communication Module** enables secure, real-time messaging within the SmartCampus ecosystem. It provides:

- Direct and group messaging for students, parents, teachers, and staff
- Broadcast announcements for school-wide or targeted communication
- File and media sharing linked to school operations
- Integrated notifications (push, email, SMS)
- Logging and auditing for compliance

The goal is to **centralize communication**, reduce reliance on external apps, and improve operational transparency.

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2. CORE OBJECTIVES

1. Facilitate **real-time one-to-one and group messaging**
2. Support **broadcasts and announcements** for school-wide updates
3. Integrate **files, media, and module-specific documents** (homework, invoices, attendance alerts)
4. Provide **role-based access and moderation**
5. Enable **audit logs for compliance and accountability**
6. Integrate with **mobile apps and parent portals**
7. Support **scalable, high-volume messaging**

3. SYSTEM ROLES & PERMISSIONS

Role	Permissions
Admin	Create/manage channels, broadcast messages, moderate content
HR / Attendance Officer	Send department-level messages, alert staff
Teacher	Communicate with students, parents, or class channels
Student	Chat with teachers or classmates (limited by class)
Parent	Chat with teachers, view announcements and child-related messages
Auditor	Read-only access to historical messages

4. MODULE ARCHITECTURE

```
In-App Communication
│
├── Chat Interface
│    ├── One-to-One Messaging
│    ├── Group Messaging (class, department, club)
│    ├── Broadcast Announcements
│    └── Threaded Replies
│
├── Notifications Engine
│    ├── Push Notifications (Mobile)
│    ├── Email / SMS Fallback
│
├── File / Media Sharing
│    ├── Documents (PDF, Excel, Word)
│    ├── Images / Videos
│    └── Attachments linked to modules (Homework, Invoice, Report)
│
├── Access Control
│    ├── Role-based Permissions
│    └── Moderation Tools
│
├── Audit & Logs
│    ├── Message timestamp
│    ├── Sender/Recipient IDs
│    └── Edit/Delete History
```

5. FUNCTIONAL WORKFLOWS

5.1 One-to-One Messaging

```
Teacher → Parent
  ↓
System verifies role & permissions
  ↓
Message encrypted & stored
  ↓
Push notification sent
  ↓
Parent reads & replies
  ↓
All messages logged
```

5.2 Group Messaging

```
Teacher creates Class 10A channel
  ↓
Students/Parents auto-added via Student Management
  ↓
Teacher posts homework instructions
  ↓
All members receive notifications
  ↓
Members reply in threaded discussions
```

5.3 Broadcast Announcements

```
Admin selects recipients (all students, parents, staff)
  ↓
Message pushed in-app
  ↓
Optional SMS/email notification
  ↓
Read receipts tracked for accountability
```

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6. MODULE INTEGRATIONS

Integration	Purpose
Student Management	Sync students and parent accounts to class channels
Staff Management	Sync teachers and staff for departmental messaging
Attendance	Notify parents/staff of absences or late arrivals
Finance	Send fee reminders, invoices, and receipts
HR	Announce staff meetings, leave notifications
Parent Portal / Mobile App	Push notifications and message access

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7. NOTIFICATIONS & ALERTS

- **Push notifications**: mobile and tablet apps
- **In-app badges**: unread message counts
- **Email fallback**: for users not actively in the app
- **Urgent alerts**: attendance, emergencies, deadlines

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8. SECURITY & COMPLIANCE

- End-to-end message encryption
- Role-based access and channel membership control
- Message retention and auditing for accountability
- Moderation tools to block inappropriate content

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9. DATA MODELS

9.1 Message Table

```
Message
- message_id
- sender_id
- recipient_id (user or group)
- content (text / file reference)
- timestamp
- read_status
- attachment_id (optional)
```

9.2 Message Group Table

```
MessageGroup
- group_id
- group_name
- members[]
- created_by
- created_at
```

9.3 Audit Table

```
MessageAudit
- audit_id
- message_id
- user_id
```

```
- action (sent, edited, deleted)
- timestamp
```

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10. SCALABILITY & RELIABILITY

- Real-time communication via **WebSockets or MQTT**
- Message queuing for **large-scale broadcasts**
- Push notifications scalable via **Firebase, OneSignal, or custom service**
- Database partitioning for long-term message storage
- Offline caching with automatic sync

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11. MOBILE & WEB INTERFACE

Features:

- One-to-one and group chats
- Threaded message replies
- File/media sharing
- Notifications (push, email, in-app)
- Quick search by person, class, or department

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12. FUTURE ENHANCEMENTS

- AI auto-categorization of messages
- Smart reply suggestions for staff/parents
- Message translation for multilingual schools
- Analytics dashboard (most active channels, response times, trends)
- Contextual messaging linked to homework, attendance, or fees

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